

Effect of Eye and Body Movement on Augmented Reality in the Manufacturing Domain

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ABSTRACT

Understanding the eye and body movements required to perform large vehicle assembly tasks is important for developing a mobile support system for mechanics, and tracking user movements with regard to the surrounding environment is critical for designing a wearable Augmented Reality (AR) systems. This poster summarizes a study measuring the eye and body movements of mechanics performing assembly activities in a live manufacturing environment. It reviews our quantitative analysis of eye movements and qualitative analysis of body movements and describes the implications of this data in terms of the feasibility and potential utility of using a mobile AR application to support manufacturing. We found that the mechanics' eye movements ranged over a slightly larger field than the eye movements reported in previous research because of constraints imposed by some body positions required in manufacturing tasks.

Keywords: Augmented reality; manufacturing; production; eye tracking; saccadic eye movement; gaze shift; heads-up display; human factors.

Index Terms: H.5.1 [Information Systems]: Multimedia Information Systems – Augmented Reality, User interface management systems J.6 [Applications]: Computer-Aided Engineering – Computer Aided Manufacturing

1 INTRODUCTION

As camera, processing, and display technologies rapidly advance, Augmented Reality (AR) applications are becoming pervasive, accessed from commercial passenger cars equipped with heads up displays (HUDs) or on small handheld electronics such as smartphones and tablets.

Interest in developing AR applications to increase productivity and reduce errors in manufacturing work has existed for more than 20 years [1]. We identified several technology barriers in current AR technology that have been slowing the development of a wearable, instructional system to support manufacturing. We performed a study to better understand these barriers with regard to the display area of AR optical see-through displays. Current optical see-through display technology provides an inlaid display as part of a set of goggles. When turned off or not illuminated, the display is clear. However, current optics manufacturing technology only allows for a 40° by 40° active area, so it does not cover the user's full field of view (more than 180° wide).

This seems like a major limitation; however previous research indicates gaze shifts generally do not exceed 15-20 degrees before being accompanied by head movements [2, 3]. Our study captured

data to determine eye and head movements as mechanics executed manufacturing tasks. Our goal was to determine if previous eyetracking research could be corroborated with large vehicle assembly mechanics, to help determine whether a display based on current technology could be useful in this domain.

2 RELATED WORK

Ong and Nee's Virtual and Augmented Reality Applications in Manufacturing text provides a nice summary of previous research, including chapters on several potential manufacturing applications based on research performed in the 1990s and early 2000s [4]. Baird (1990) was among the first to investigate the feasibility of using AR technology in a manufacturing environment [1]. More recent related work includes Steve Henderson's Augmented Reality for Maintenance and Repair (ARMAR) project, which investigates engineering and interaction challenges within an AR based maintenance system [5].

3 EXPERIMENT APPROACH

Our study was performed at a large vehicle assembly facility as mechanics performed their normal assembly activities, or "work orders," while wearing the eye-tracking hardware built by the experiment team.

3.1 Experiment Setup

Our mobile eye-tracking system included a head-mounted eye-tracking unit and a small laptop computer with software to capture and analyze the video.

The hardware design followed the same basic principles as described in [6]. Due to the nature of the manufacturing environment we could not cut away the plastic portion of the safety goggles; this introduced some glare into the picture (due to the LEDs and ambient lighting). However, tracking was still good through the reflective plastic medium.

We used ITU Gaze Tracker software to track eye movement using an in-facing camera. An outfacing, "first-person" perspective camera provided reference information about the task as well as information about body posture, movement throughout the work area, and use of tools and reference material. Video from both cameras was recorded using a screen capturing program to record both visual outputs at the same time. This allowed us to simultaneously observe the eye position and the first-person perspective of the subject while analyzing our data.

3.2 Experiment Execution

Six mechanics outfitted with the experimental setup performed their normal work orders in a large open area with individual work areas measuring approximately 10'x10' (see Figure 1). One data set was collected for each mechanic for each work order. These work orders ranged from thirty to ninety minutes and included multiple tasks, such as attaching bolts, clamps, fittings, etc. to a large metal subassembly. The mechanic frequently moves around during the execution of a work order to find parts and tools as well as refer to materials about the work being performed.

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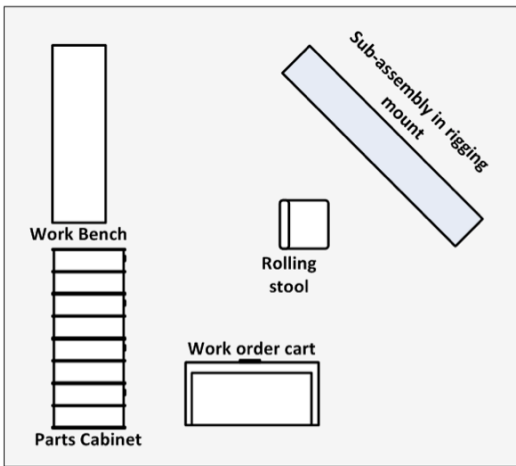


Figure 1: Work area layout

Based on qualitative analysis of the first person video, task type and mechanic movement affected eye movements during the sessions.

4 ANALYSIS

To understand the effect of the eye movements on the feasibility of having mechanics use a mobile AR system to support manufacturing work, the foci of the data analysis for this study were:

- Understanding the ranges of physical eye movements of large vehicle assembly mechanics.
- Measuring the movement of mechanics eyes throughout the execution of work orders. (Movements of less than 2° were not noted due to possible errors in the analysis due to camera resolution and image processing.)
- Computing the percentage of eye movements below 20° in each direction (40° total).

4.1 Eye Movement Analysis

Initially, the raw data was plotted visually and data points beyond the physical limitations of the eye were labelled as system misrecognitions of the pupil. They were filtered out during subsequent calculations. The resultant graph provided insight into the dense, center-focused nature of eye movement through the task.

For analysis, pupil distance from center was calculated and grouped into 10° increments from the center point of the eye (e.g., 0° - 10° , 10° - 20° , 20° - 30° , etc.). The focus of the study was on the currently available 40° field of view hardware, so the results were further aggregated to less than and greater than 40° . The movement was also broken down by axis for further analysis.

In the work environment, it was difficult to properly calibrate the pupil detection system; therefore, results were calculated at the maximum and minimum calibration distance to allow analysis of both extremes.

Previous research has found the vast majority of eye movement is less than 20° due to accompanying head movement when objects outside that range need to be viewed [2, 3]. Our study resulted in similar findings, but the number of gaze shifts outside 20° was slightly higher than found in previous research. Still, the subjects' gaze averaged at least 67% within the active area on the horizontal plane (calculated in the minimum case). Overall eye gaze (horizontal and vertical) was within the active area at least 48% of the time (again the minimum case).

To better understand the deviations from earlier research, we analyzed the eye movement results in light of the body posture and task analysis from the captured video. Generally, eye movement remained within the active area while the mechanic was focused on the hull during assembly, which is when the mobile support system would be delivering the most critical information. We believe this indicates that mechanics would benefit from an AR app running on a mobile, heads-up display-based system designed to support them in performing manufacturing tasks.

4.2 Body Posture Analysis

The outward-facing camera of the eye-tracking hardware provided a unique "first-person" perspective of the mechanic, offering clues about body positions required to perform work orders. Overall analysis of the video and observation during the study showed the mechanics focus was largely on just a few items: the hull piece being assembled, the part container, small parts used for the work order, handheld power tools, and reference material. The instructions often dictated the eye and body movement and heavily impacted the measured gaze shift.

For instance, Session 898 yielded the largest gaze shift outside of the target 40° and deviated most from the previous research. The video analysis revealed that early in the work order the mechanic operated a handheld grinder on a small assembly part for several minutes. The mechanic performed the task while standing and holding the grinder at waist-level, looking down throughout the work order.

Conversely, the mechanic in Session 739 yielded the smallest gaze shift outside 40° . The video analysis showed the piece on which the mechanic was working was already located at eye level in the hull rigging.

5 CONCLUSIONS AND RECOMMENDATIONS

Although lying slightly outside the results of previous research, the majority of the gaze shifts we measured while mechanics performed assembly tasks fell within the 40° field of view supported by the "active area" of current AR goggles. These results suggest that current head mounted display technology would benefit the manufacturing domain; however, future display technology expanded to 60° or greater would have a greater impact on a mechanic's assembly tasks. We recommend a larger study that decomposes work orders to the tasks level in the work environment in order to better understand the limitations of AR systems in this domain.

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